# 005 BE Practical-1.

HT NO: 2203A51360

BATCH NO:29

QUESTION:

To develop and perform unit testing of smart contracts using the Truffle framework

## Objective:

Implement a simple Solidity Smart Contract and perform UNIT Testing using Truffle to verify contract functionality.

Problem Statement (Equivalent to Given Cipher Practical)

Develop a basic Personal Portfolio Smart Contract that stores and retrieves user profile data, and validate its behavior using Truffle unit tests.

## Requirements

Development Environment

• Install Node.js

• Install Truffle Framework

• Install Ganache (Local Blockchain)

• Use VS Code with:

o Solidity Extension

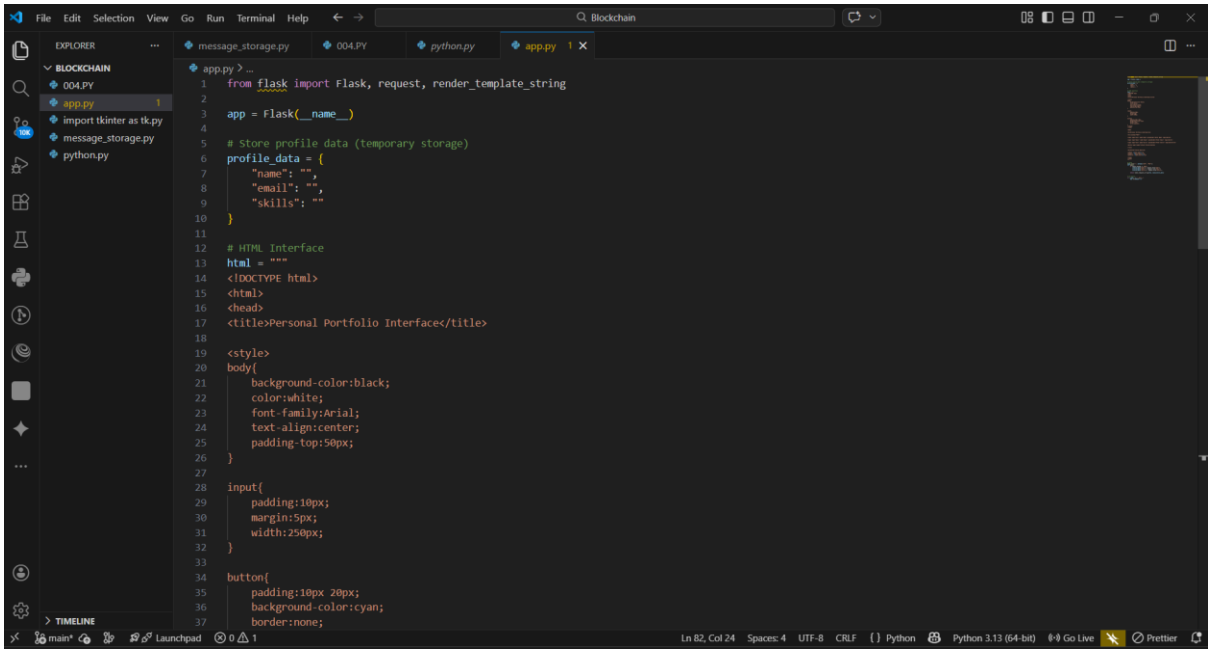
o JavaScript Extension

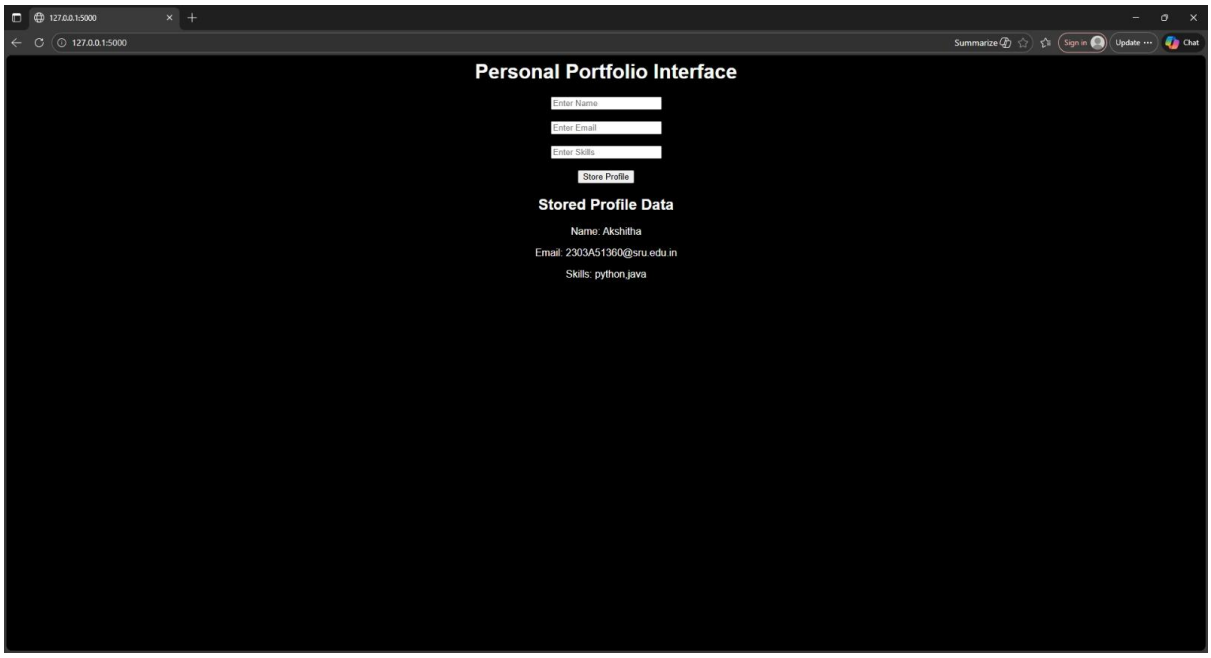
CODE:

```
from flask import Flask, request, render_template_string
app = Flask(__name__)
# Store profile data (temporary storage)
profile_data = {
    "name": "",
    "email": "",
    "skills": ""
}
# HTML Interface
html = """
<!DOCTYPE html>
<html>
<head>
<title>Personal Portfolio Interface</title>
<style>
body{
    background-color:black;
    color:white;
    font-family:Arial;
    text-align:center;
    padding-top:50px;
}
input{
    padding:10px;
    margin:5px;
    width:250px;
}
```

```
button{
    padding:10px 20px;
    background-color:cyan;
    border:none;
    cursor:pointer;
}
</style>
</head>
<body>
<h1>Personal Portfolio Interface</h1>
<form method="POST">
<input type="text" name="name" placeholder="Enter Name" required><br>
<input type="email" name="email" placeholder="Enter Email" required><br>
<input type="text" name="skills" placeholder="Enter Skills" required><br><br>
<button type="submit">Store Profile</button>
</form>
<h2>Stored Profile Data</h2>
<p>Name: {{data.name}}</p>
<p>Email: {{data.email}}</p>
<p>Skills: {{data.skills}}</p>
</body>
</html>
"""
# Route
@app.route("/", methods=["GET", "POST"])
def home():
    if request.method == "POST":
        profile_data["name"] = request.form["name"]
        profile_data["email"] = request.form["email"]
```

```
        profile_data["skills"] = request.form["skills"]
    return render_template_string(html, data=profile_data)
# Run Server
if __name__ == "__main__":
    app.run(debug=True)
```




OUTPUT: